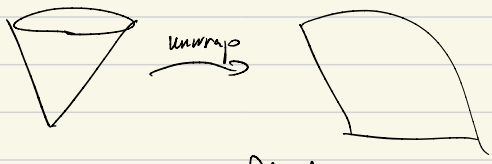
more 1st fund. form examples 2

1) plane

2) cone $\sigma(r,\theta) = (r\cos\theta, r\sin\theta, r) \longrightarrow ds^2 = 2dr^2 + r^2 d\theta^2$

note: the cone is "flat"



Applying standard polar coordinates to the flattened picture since

$\sigma(r,\theta) = \left(\frac{r}{\sqrt{2}}\cos(\sqrt{2}\theta), \frac{r}{\sqrt{2}}\sin(\sqrt{2}\theta), \frac{r}{\sqrt{2}}\right)$ (same cone $\{z=r\}$)

$\longrightarrow ds^2 = dr^2 + r^2 d\theta^2$ Euclidean!

3) hyperboloid $\{r^2 = z^2 + 1\}$

std. $\sigma(t,\theta) = (\cosh t\cos\theta, \cosh t\sin\theta, \sinh t)$

$ds^2 = dt^2 + \cosh^2 t\, d\theta^2$

ruled surface let $\gamma(\theta) = (\cos\theta, \sin\theta, 0)$

$\sigma(t,\theta) = \gamma(\theta) + t(\gamma'(\theta) + e_3)$ Check: $r^2 = 1+t^2$, $z = t$

$ds^2 = dt^2 + 2\,dt\,d\theta + (1+t^2)\,d\theta^2$

3) catenoid $\{r = \cosh z\}$

$\sigma(z,\theta) = (\cosh z\cos\theta, \cosh z\sin\theta, z)$

$\Longrightarrow ds^2 = \cosh^2 z\,(dz^2 + d\theta^2)$

Maps between surfaces

def.: $\psi : S \to \tilde{S}$ is smooth if it's representation in local coord's is smooth for all patches

The differential of ψ is a family of linear maps

$$d\psi_p : T_pS \longrightarrow T_{f(p)}\tilde{S}$$

defined by $d_p\psi(X)f := X(f\circ\psi)$ for $\begin{cases} X \in T_pS \\ f \in C^\infty(\tilde{S}) \end{cases}$

In coord's (u^1, u^2) for S, (v^1, v^2) for $\tilde{S}$, we can write ψ as

$$\psi(u^1, u^2) = (v^1(u^1, u^2), v^2(u^1, u^2))$$

$$d\psi(\partial_i)f = \frac{\partial}{\partial u^i} f(v^1, v^2)$$

$$= \frac{df}{\partial v^j}\frac{\partial v^j}{\partial u^i}$$

$$\therefore d\psi : \partial_i \longmapsto \frac{\partial v^j}{\partial u^i}\tilde{\partial}_j \qquad \text{(the chain rule!)}$$

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def.: ψ is an local isometry if lengths of curves are preserved

$$\ell(\gamma) = \ell(\psi\circ\gamma) \quad \text{for all } \gamma : I \to S$$

Since $\psi\circ\gamma = d\psi(\dot\gamma)$ by def of $d\psi$

local isom. $\iff$ $\|d\psi(\dot\gamma)\| = \|\dot\gamma\|$ for all curves

We can express this in terms of a pullback of the fundamental form $\tilde{g}$ on $\tilde{S}$

define: $\psi^*\tilde{g}(X,Y) := \tilde{g}(d\psi(X), d\psi(Y))$ for $X, Y \in T_pS$

Then $\|X\|^2 = \|d\psi(X)\|^2$ for all X means

$$g(X,X) = \psi^* \tilde{g}(X,X)$$

Because these are real inner products, this is equiv. to

$$g(X,Y) = \psi^* \tilde{g}(X,Y)$$

Thm ψ is a local isometry $S \to \tilde{S}$ if

$$\psi^* \tilde{S} = g.$$

We write this as

$$dA = \sqrt{\det g}\, du^1 du^2$$

examples

1) cylinder $r(z,\theta) = (\cos\theta, \sin\theta, z)$

$$ds^2 = dz^2 + d\theta^2, \quad \det g = 1 \Longrightarrow dA = dz\, d\theta$$

2) sphere in cylindrical coordinates $\{r^2 + z^2 = 1\}$

$$r(z,\theta) = (\sqrt{1-z^2}\cos\theta, \sqrt{1-z^2}\sin\theta, z)$$

$$ds^2 = \frac{1}{1-z^2}dz^2 + (1-z^2)d\theta^2$$

$$\det g = 1$$

$dA = dz\, d\theta$ (same as for cylinder!)